

Accuracy-Centered Baseline Comparison Results

Baseline comparison report

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Purpose

This report presents two baseline comparisons for the proposed MFG posterior estimator. **Scheme A** compares Algorithm 1 with B1–B3 at the same single-pass/local-update level. **Scheme B** compares Algorithm 2 with multi-pass residual-safeguarded refinements built from B1–B3 candidate generators. All runs use the false-confidence Experiment 2C protocol, the same initial pose, and the same seeds [42, 43, 44, 45, 46]. All reported methods are evaluated under the same protocol and measurement design.

Methods and References

B0 Initial is the common initial/prior-mode pose. **B1 Lie-GN/LM MAP** is a conventional damped nonlinear least-squares MAP estimator on $SO(3) \times \mathbb{R}^3$, using Gauss–Newton/Levenberg–Marquardt ideas [1, 2, 3] and right-perturbation Lie-group coordinates [4]. **B2 Tangent Gaussian/Laplace** applies the one-shot local quadratic/Laplace posterior mode at the initial linearization point [5, 4]. **B3 Decoupled local Gaussian** is not a separate standard estimator; it is a targeted ablation of the paper’s coupled posterior structure, obtained by zeroing the rotation–translation cross-information block before solving the same local quadratic update. **Algorithm 1** is the paper’s single-pass closed-form MFG posterior update. **Algorithm 2** is the paper’s residual-safeguarded multi-pass MFG Bayesian refinement.

Experimental Setting

The protocol is the false-confidence prior-mismatch case with $(\beta, c_\kappa, c_\Lambda) = (2, 2, 2)$ and $K = 10$ sparse-baseline wrench measurements. The base initial perturbation is specified by the experimental protocol, corresponding to the scaled initial errors $e_R = 0.2631$ rad and $e_p = 14.59$ mm. The prior confidence is $\kappa_0 = 120$ and $\Lambda_0 = \text{diag}(12000, 12000, 12000)$. The wrench-noise levels are $\sigma_\tau = 0.10$ and $\sigma_f = 0.70$. Scheme A uses one local update for each method. Scheme B uses at most $T_{\max} = 20$ outer re-centering steps for B1–B3 and the 20-step Proposed Alg. 2 runs. For B1, the maximum number of single-pass GN/LM iterations in Scheme A is 1; in the table below Scheme A uses the single-step B1 row, while Scheme B uses one B1 candidate step per outer refinement iteration.

Scheme A: Single-Pass / Local-Update Comparison

Positive Δe_R and Δe_p mean reduction relative to the identical initial error. The comparison is therefore about estimation accuracy, while runtime is kept as a secondary diagnostic.

Method	n	init e_R	final e_R	Δe_R	init e_p (mm)	final e_p (mm)	Δe_p	ρ
B0-Initial	5	0.263 ± 0.000	0.263 ± 0.000	0.0 ± 0.0	14.59 ± 0.00	14.59 ± 0.00	0.0 ± 0.0	15.36 ± 0.43
Alg1-MFG-single-pass	5	0.263 ± 0.000	0.302 ± 0.008	-14.6 ± 2.9	14.59 ± 0.00	9.19 ± 1.18	37.0 ± 8.1	14.73 ± 1.21
B1-LieGN-LM-single-step	5	0.263 ± 0.000	0.288 ± 0.023	-9.5 ± 8.8	14.59 ± 0.00	11.47 ± 3.02	21.4 ± 20.7	14.67 ± 0.75
B2-Tangent-Gaussian-Laplace	5	0.263 ± 0.000	0.302 ± 0.008	-14.6 ± 2.9	14.59 ± 0.00	9.29 ± 1.15	36.3 ± 7.9	14.88 ± 1.02
B3-Decoupled-local-Gaussian	5	0.263 ± 0.000	0.336 ± 0.011	-27.7 ± 4.3	14.59 ± 0.00	9.19 ± 1.18	37.0 ± 8.1	14.63 ± 1.26

Scheme A Runtime

Method	time (s)	iterations	accepted updates	fail. %
B0-Initial	2.15 ± 0.06	0.0 ± 0.0	–	0.0
Alg1-MFG-single-pass	4.39 ± 0.21	1.0 ± 0.0	–	0.0
B1-LieGN-LM-single-step	4.31 ± 0.09	1.0 ± 0.0	–	0.0
B2-Tangent-Gaussian-Laplace	4.34 ± 0.13	1.0 ± 0.0	–	0.0
B3-Decoupled-local-Gaussian	4.36 ± 0.09	1.0 ± 0.0	–	0.0

Scheme B: Multi-Pass Refinement Comparison

For B1–B3, each outer step recomputes the local information at the current nominal pose, proposes a method-specific local candidate, tests relaxed step lengths $\alpha \in \{1, 0.5, 0.25, 0.1\}$, and accepts only if the recomputed whitened residual merit decreases. The returned estimate is the best accepted nominal pose, matching the role of the accepted re-centering pose in Algorithm 2.

Method	n	init e_R	final e_R	Δe_R	init e_p (mm)	final e_p (mm)	Δe_p	ρ
B0-Initial	5	0.263 ± 0.000	0.263 ± 0.000	0.0 ± 0.0	14.59 ± 0.00	14.59 ± 0.00	0.0 ± 0.0	15.36 ± 0.43
B1-LieGN-LM-refinement	5	0.263 ± 0.000	0.322 ± 0.039	-22.4 ± 14.9	14.59 ± 0.00	8.11 ± 2.13	44.4 ± 14.6	12.02 ± 2.48
B2-Tangent-Gaussian-refinement	5	0.263 ± 0.000	0.311 ± 0.023	-18.2 ± 8.8	14.59 ± 0.00	8.56 ± 1.50	41.4 ± 10.3	12.75 ± 1.94
B3-Decoupled-Gaussian-refinement	5	0.263 ± 0.000	0.395 ± 0.043	-50.2 ± 16.3	14.59 ± 0.00	6.65 ± 0.90	54.4 ± 6.2	10.64 ± 1.46
Proposed-Alg2-refinement	5	0.263 ± 0.000	0.246 ± 0.009	6.6 ± 3.2	14.59 ± 0.00	1.08 ± 0.16	92.6 ± 1.1	6.78 ± 0.51

Scheme B Runtime

Method	time (s)	iterations	accepted updates	fail. %
B0-Initial	2.15 ± 0.03	0.0 ± 0.0	–	0.0
B1-LieGN-LM-refinement	51.23 ± 20.01	5.0 ± 1.9	4.0 ± 1.9	0.0
B2-Tangent-Gaussian-refinement	48.99 ± 4.92	4.8 ± 0.4	3.8 ± 0.4	0.0
B3-Decoupled-Gaussian-refinement	60.08 ± 23.72	5.4 ± 1.8	4.4 ± 1.8	0.0
Proposed-Alg2-refinement	687.29 ± 33.35	20.0 ± 0.0	20.0 ± 0.0	0.0

Main Observations

Scheme A should be read only as a single-pass/local-update comparison. Algorithm 1 and the one-shot local Gaussian variants mainly reduce the translation error in this false-confidence case, while all one-shot corrections show negative mean rotation-error improvement. This does not support an overclaim that a single local update fully fixes the pose; it records the limited one-pass behavior under the common initialization.

In Scheme B, B1–B3 are allowed the same outer budget $T_{\max} = 20$ as a multi-pass refinement budget, but the residual safeguard often stops them early when no tested local candidate further decreases the recomputed whitened residual merit. They reduce translation error and residual compared with the initial pose, but their mean rotation error increases in this protocol. The Proposed Algorithm 2 row is the only row in this table that reduces both mean rotation and mean translation error, although it is also substantially slower. Thus the defensible revision claim is about accuracy of the coupled MFG Bayesian refinement in this prior-mismatch experiment, not about runtime.

Interpretation

Scheme A addresses single-pass fairness: Algorithm 1 is compared only with one-shot or one-step local baselines. Scheme B addresses refinement fairness: B1–B3 receive the same kind of repeated re-centering and residual safeguard as Algorithm 2, but use their own local candidate generators. Thus any advantage of Proposed Algorithm 2 in Scheme B is not merely due to running multiple outer iterations; it reflects the posterior candidate construction and coupled MFG update used by the proposed refinement.

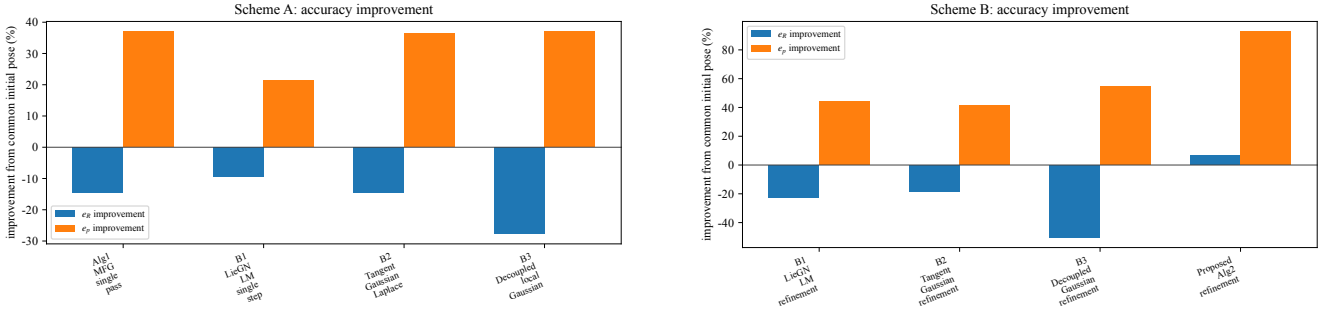


Figure 1: Mean accuracy improvement from the identical initial pose for Scheme A and Scheme B.

Reproducibility Notes

Raw records and summaries are written in `results/`. Figures are written in `figures/`. This report was generated with `run_protocol_accuracy_baselines.py` using `--seeds 42 43 44 45 46 --max-iter 1 --T-max 20`. All entries correspond to the same false-confidence Experiment 2C protocol and sparse measurement design.

References

- [1] K. Levenberg, “A method for the solution of certain non-linear problems in least squares,” *Quarterly of Applied Mathematics*, vol. 2, no. 2, pp. 164–168, 1944.
- [2] D. W. Marquardt, “An algorithm for least-squares estimation of nonlinear parameters,” *SIAM Journal on Applied Mathematics*, vol. 11, no. 2, pp. 431–441, 1963.
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- [4] T. D. Barfoot, *State Estimation for Robotics*. Cambridge, U.K.: Cambridge University Press, 2017.
- [5] L. Tierney and J. B. Kadane, “Accurate approximations for posterior moments and marginal densities,” *Journal of the American Statistical Association*, vol. 81, no. 393, pp. 82–86, 1986.